Boltzmann eventually triumphed in the fight against Ostwald and the adherents of Energetics, as it had been self-evident to me that he would, in view of all that I have just mentioned. The basic difference between the conduction of heat and a purely mechanical process became universally recognized. This experience gave me also an opportunity to learn a fact—a remarkable one, in my opinion: A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die, and a new generation grows up that is familiar with it.

Otherwise, the controversies just mentioned held comparatively little interest for me, as they could not be expected to produce anything new. My attention, therefore, was soon claimed by quite another problem, which was to dominate me and urge me on to a great many different investigations for a long time to come. The measurements made by O. Lummer and E. Pringsheim in the German Physico-Technical Institute, in connection with the study of the thermal spectrum, directed my attention to Kirchhoff's Law, which says that in an evacuated cavity, bounded by totally reflecting walls, and containing any arbitrary number of emitting and absorbing bodies, in time a state will be reached where all bodies have the same temperature, and the radiation, in all its properties including its spectral energy distribution, depends not on the nature of the bodies, but solely and exclusively on the temperature. Thus, this so-called Normal Spectral Energy Distribution represents something absolute, and since I had always regarded the search for the absolute as the loftiest goal of all scientific activity, I eagerly set to work. I found a direct method for solving the problem in the application of Maxwell's Electromagnetic Theory of Light. Namely, I assumed the cavity to be filled with simple linear oscillators or resonators, subject to small damping forces and having different periods; and I expected the exchange of energy caused by the reciprocal radiation of the oscillators to result, in time, in a stationary state of the normal energy distribution corresponding to Kirchhoff's Law.

This extended series of investigations, certain ones of which could be verified by comparisons with known observational data, such as the measurements of damping by V. Bjerknes, resulted in establishing the general relationship between the energy of an oscillator having a definite period, and the energy radiation of the corresponding spectral region in the surrounding field when the exchange of energy is stationary. From this there followed the remarkable result that this relationship is absolutely independent of the damping constant of the oscillator—a circumstance which was very pleasing and welcome to me, because it permitted the entire problem to be simplified, by substituting the energy of the oscillator for the energy of the radiation, thus replacing a complicated structure possessing many degrees of freedom, by a simple system with just one degree of freedom.